Can Humans Think Machines Think?

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This is a description of how the use of computers proved to be valuable in teaching a course on minds and machines. The first half of the course dealt with traditional problems in philosophy of mind as they apply to humans: mind/body distinctions, the problem of other minds, and related underlying issues—behaviorism, reductionism, causation. This led up to a discussion of Turing's interesting treatment of the question, "Can machines think?"

While this is a question that students find intriguing, it can be a difficult one to get students to treat philosophically. Many science fiction fans and computer science majors are convinced that machines can do anything, without really understanding how some of the feats they envision could be accomplished. And some students with anti-technology prejudices proclaim that machines are not human, assuming that such a claim settles the matter. The philosophical issues can be quickly buried beneath a trading of prejudices.

We found that an effective way to encourage philosophical discussion of the mind-machine problem was to give class participants some exposure to "intelligent" machines. Students were given the opportunity to converse with ELIZA, a program which simulates a psychotherapist, and to play a simple game with a computer program which "learns."

Acquaintance with these programs provided students with some of the background necessary to critically discuss the question, "Can machines think?" Students who had previously claimed that the answer is "No" were forced to think a little more carefully about what it is they think machines cannot do that people can. Students who had previously answered "Yes" had a better idea of the problems and difficulties that stand in the way of arguing for such a position. Both groups found that a number of the philosophical issues discussed earlier in the semester became a little less abstract—even students who were not at all worried about using the Argument from Analogy to conclude that other *humans* have minds realized that the same sort of argument might apply to robots.

In what follows we will discuss how experience with computer programs can be used to give students a better understanding of the philosophical issues

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involved in the question "Can machines think?" We will describe how the programs were introduced to the class, and some of the questions that were raised about the programs after students had an opportunity to work with them. In our final section, we will discuss the mechanics of setting up the computer sessions. (Very little skill in computer use is needed on the part of the instructor.)

I

In 1950 Alan Turing offered a substitute for the question "Can machines think?" He first described a game with three players: A, a woman; B, a man; and C, the interrogator. The object of the game for C is to guess which of the other two is the woman. Player C may ask each of them questions, but they are not in the same room as C and C can communicate with A or B only by teletypewriter or through an intermediary, so that C cannot get information about their physical characteristics. The object of the Imitation Game for both A and B is to convince C that she/he is the woman. The question to substitute for "Can machines think?" is: "What will happen when a machine takes the part of A in the game, where the object for C is to guess which one, A or B, is the *human*? Will C decide wrongly as often when the game is played in this version as when the game is played with a female and male?" Turing predicted that within fifty years the answer to the last question would be "Yes."

Turing argued that objections to the possibility of a computer thinking are either traditional philosophy of mind arguments which lead to skepticism about the possibility of other humans' thinking as well, or misrepresentations of the actual abilities of machines. In support of his prediction, Turing suggested we imagine a computer constructed with only the abilities of a human infant. Given the same sort of environment as a child—the same stimuli, rewards, and punishments—the computer could be made to change its responses as a result of those stimuli and acquire the same sort of abilities and experiences as those of a human adult. If we can imagine that a computer could be so constructed, then, Turing thought, we would be able to agree that the computer could learn to play the Imitation Game as well as a human.

The arguments for and against the possibility of machines thinking involve conjectures about future or possible machines which are based on claims about current machine capabilities. For example, arguments that machines could never think frequently use premises of the form:

> Machines could never do X.
> X is essential for thinking.

where X is some sort of behavior currently attributed only to humans. The first premise is a somewhat risky claim, for it may expose one to the ridicule in the limerick:

> Philosophers gravely expound
> Metaphysical concepts profound
> They argue all night
> On the meaning of flight
> While the Wrights get a plane off the ground.[1]

New situations often inspire extensions or variations in old concepts. A conceptual analysis of X which rules out certain applications may not recognize the fluid character of concepts which allows them to have new applications.

On the other hand, arguments that machines *could* think may use premises of the form:

> Machines cannot yet do X, but they can do Y.
> The difference between machines doing X and machines doing Y is only a matter of technology.

Similar claims have been made when perpeutal motion, or speeds faster than light have been in the place of "X," and so we should be suspicious of arguments of this form.

Some understanding of the actual abilities and accomplishments of machines is required to evaluate these speculations about what possible machines could or could not do. Information about what current machines can do and how they do it is important not only for discussing the "Can machines think?" question, but for evaluating a number of other claims that are becoming of increasing concern with the growing use of computers. Computer theft, computer invasion of privacy, or even computer management of battle decisions are only a few of the issues that require some understanding of computer capabilities. But simply providing such information through lectures and reading turns out not to be a very effective means of teaching. For students unfamiliar with computers, whose paradigms of machines are automobiles and refrigerators, being told that computers have discovered new mathematical proofs, made mistakes, learned from experience, or responded in varied and novel ways to the same input, is about as unhelpful as simply being told that machines can think. Either this information will have to be accepted on faith or it will be rejected because it seems incomprehensible that a machine could do such things: It will be thought that "prove," "learn," and "respond in novel ways" are being used in an extended or metaphoric sense, or that all the intelligent aspects of the computer's behavior are nothing more than the results of a program of an intelligent human. Even people who are familiar with computers often view them as mere calculating tools and have a poor understanding of the more complex tasks machines can perform.

One way to make claims about machine capabilities comprehensible to students would be to give them a detailed account of how computers operate and of the programs available. But even if time for this were available, the vast differences in knowledge about computers among the students made this approach unacceptable. Fortunately, a comprehensible explanation can usually be given without filling in all the details. For example, although one knows almost nothing about the mechanisms involved, the elimination of a pain can be satisfactorily explained by the taking of aspirin, or reduced pollution explained by the use of emission control devices. And in many cases, a demonstration of these effects will contribute more to one's understanding than any amount of discussion. Therefore, instead of giving a rigorous account of how a computer

works, we gave course participants a chance to work with some actual programs in order to give them a better understanding of computer abilities.

The programs we selected were used because their principles of operation are relatively simple and easy to see from a demonstration; they are not representative of the latest developments in artificial intelligence. They were not intended to represent the working of the human mind, nor were they intended to demonstrate that computers do think. Nonetheless, experience with these programs gave students a better perspective from which to evaluate Turing's claims. The programs have implications for a wide variety of philosophical issues, helped to eliminate inaccurate stereotypes about machines, and provided a basis for original thinking about the problems of minds and machines that might not have been achievved through lecture-discussion format alone.

II. *A Conversation with* ELIZA

Less than twenty years after Turing's prediction, Joseph Weizenbaum created a computer program called ELIZA. One version of ELIZA was designed to respond to teletyped English messages the way a Rogerian psychotherapist would respond to a client. Rogerian therapy, after Carl Rogers, encourages clients to find their own solutions and use their own values and avoids directing clients toward any particular set of values or solutions. Consequently, appropriate responses from such a therapist are likely to be requests to explain things more, or rephrasings and reorganizings of the client's previous remarks, containing little new information which might impose outside values and views on the client. It is this lack of new information which makes such responses comparatively easy to simulate.

Soon the story flashed around computer circles that this version of the ELIZA program had passed Turing's test. It did not pass Turing's test, but it came close, much closer than Turing and other researchers in artificial intelligence would have expected for such a simple program.

A series of interactions with ELIZA show its limitations and provide insight into the nature of its responses. ELIZA has a repertoire of stock contexts in which to embed the input sentences, rules to make some simple transformations, a set of cue words which can trigger a specific sort of response. With these simple but clever devices, the program can make apparently coherent and appropriate responses to an enormous number of sentences. Here are some excerpts of the conversations with ELIZA from the course.

ELIZA: HOW DO YOU DO, PLEASE TELL ME YOUR PROBLEM (This is ELIZA's opening line.)

Student 1: I AM AFRAID OF HEIGHTS

E: IS IT BECAUSE YOU ARE AFRAID OF HEIGHTS THAT YOU CAME TO ME? (The program will change I AM to YOU ARE and YOU to I and embed the previous comments in a context such as: IS IT BECAUSE_____THAT YOU CAME TO ME?)

S1: YES, I WANT TO GET OVER THIS FEAR

E: WHAT WOULD IT MEAN TO YOU IF YOU GOT TO GET OVER THIS FEAR? (A change of I WANT to YOU GOT to embed in WHAT WOULD IT MEAN TO YOU IF_____)
S1: I WOULD BE ABLE TO FLY IN A JET
E: DO YOU SAY YOU WOULD BE ABLE TO FLY IN A JET FOR SOME SPECIAL REASON? (The repertoire of embeddings with pronoun change continues from here.)
S1: YES, I WANT TO FLY TO COLORADO
E: WHY DO YOU WANT TO FLY TO COLORADO?
S1: I WANT TO GO SKIING
E: SUPPOSE YOU GOT TO GO SKIING SOON
S1: I WOULD BE TOO AFRAID TO FLY
E: THAT'S QUITE INTERESTING (One of several noncommital phrases. Not really appropriate here.)
S1: I NEED TO GET OVER THIS FEAR
E: WHAT IF YOU NEVER GOT TO GET OVER THIS FEAR? (WHAT IF YOU NEVER GOT has been substituted for I NEED.)
S1: I WOULD BE VERY DEPRESSED
E: YOU SAY YOU WOULD BE VERY DEPRESSED

And so on. Often, though, ELIZA's responses are less realistic. Another student, after trying to discuss a "problem" of not submitting papers on time, suddenly resorted to typing in numbers. ELIZA continued to respond with noncommital phrases:

Student 2: 14706
E: DOES THAT HAVE ANYTHING TO DO WITH THE FACT THAT YOUR LAST TWO PAPERS WERE LATE? (From a program instruction to pick up on an earlier input.)
S2: 306
E: I AM NOT SURE THAT I UNDERSTAND YOU FULLY
S2: YOU SHOULDN'T
E: YOU'RE NOT REALLY TALKING ABOUT ME—ARE YOU? (This might seem to be an appropriate response if the previous input by the student were taken out of context. It is often ELIZA's comments taken as a whole series that make the least sense. The program does not consider ELIZA's earlier comments, or the sense of the whole dialogue. ELIZA's responses are based on a single input—usually but not always the most recent—from the teletypewriter.)
S2: 722
E: PLEASE GO ON
S2: 9
E: DO YOU FEEL STRONGLY ABOUT DISCUSSING SUCH THINGS?
S2: 5237
E: WHAT ARE YOUR FEELINGS NOW?
S2: I LOVE YOU
E: PERHAPS IN YOUR FANTASY WE LOVE EACH OTHER (pretty good)
S2: GET YOUR HAND OFF MY THIGH

But ELIZA wins the round with:

E: DOES THAT SUGGEST ANYTHING ELSE WHICH BELONGS TO YOU?

Although ELIZA's responses are often remarkably appropriate, such as:

S3: TALKING TO YOU IS LIKE TALKING TO A BRICK WALL
E: WHAT RESEMBLANCE DO YOU SEE?
S3: DID YOU UNDERSTAND WHAT I SAID?
E: YOU SAY DID I UNDERSTAND WHAT YOU SAID

they can also be ungrammatical and unintelligible, such as:

S3: I DO NOT WANT TO DIE UNITL I AM OLD ENOUGH
E: PERHAPS IN YOUR FANTASY WE DO NOT WANT TO DIE UNTIL EACH OTHER (probably a program execution error)
S3: YOU SEEM TO HAVE MORE PROBLEMS THAN I DO
E: YOU SEEM TO NEED TO THINK ME

If several rounds of the Imitation Game were played, it is unlikely that ELIZA would fool a skilled interrogator. Repeated responses are recognizable and absurdities can be generated.[2]

The experience with ELIZA was supplemented by readings and discussions on philosophic topics in artificial intelligence.

More recent computer programs for natural language such as Colby's PARRY have more semantic connections and consequently produce a more convincing dialogue. Any lack of continuity and coherence of the dialogue as a whole enhances rather than detracts from this program because PARRY is (an imitation of) a paranoid schizophrenic. Significantly, Winograd's SHRDLU, which converses about an "actual world" of blocks and pyramids, does not have trouble carrying on a connected, continuous conversation. Although SHRDLU's world and topics of conversation are very limited, it is far more coherent and intelligible than the programs which rely entirely on syntactic rules and semantic connections among words. SHRDLU also has associations between words and the things in its world and is able to ask questions about unclear statements and use new information about its world obtained from the conversation in constructing its own responses.

The question "Can machines think?" naturally leads to the question "What is the nature of thinking?" Experience with computers provided the class with a new way of asking this question. For example, How could one build something that could think? What other properties would it have to have? What would it be able to do? In the spirit of these questions we discussed the relation of language to thought. This topic is additionally relevant because of the number of primates learning language and because of questions of animal rights.

The ELIZA program's success does not show that computers can think, but it may show that conversational dexterity is not the evidence we use in deciding what things can think. Even though ELIZA could not pass Turing's test in the long run, that simple system of stock answers did come up with appropriate responses to an enormous range of remarks. Yet we do not believe that a computer with the ELIZA program thinks, even a little bit. As for humans, we may believe that many of them think differently from the way we do, that they do

not think much, etc., but however repetitious, superficial, or inappropriate their use of language, we do not doubt that they think at all. Therefore, the class concluded, perhaps language use is not the criterion of thought.

Next the class discussed Michael Arbib's observation:

> Artificial intelligence (AI) may be defined as the area concerned with programing computers to behave in a way that one would call intelligent—if one did not know that a computer could do it! Unfortunately, once one has explained how a machine can be programed to perform some process, it is hard for a human to regard the process as requiring intelligence.[3]

The class considered whether the belief that humans think and (existing) machines do not was just prejudice, protein chauvinism with no rational basis. But even those course participants who claimed they liked machines better than people concurred that a computer with the ELIZA program did not think.

Arbib goes on to say that intelligence may be:

> ...a plexus of properties which, taken one at a time, may be little cause for admiration, but any sizable portion of which will yield behavior that we would label as intelligent.[4]

In a similar vein, Gunderson has argued that thinking could not be reduced to net results, i.e., performance on a single task, but required a whole range of abilities. Turing undoubtedly would have agreed, but thought the Imitation Game *would* be an adequate test of a whole range of abilities. The class discussed these hypotheses and then considered a more specific suggestion: Perhaps the ability to *learn* is important in the attribution of thinking, and the ability to perform is not enough. Turing had assumed that linguistic ability was the criterion for thinking. To support his claim that a machine could play the Imitation Game, he argued that a machine could learn like a human child. Perhaps learning is a criterion for thinking and linguistic ability is only a sign (at best) that learning has taken place. It is in the development of this topic that the next experience with the computer takes place.

III. *Machine Learning*

One thing we can take for granted about humans and some other living organisms, but not about machines, is that many of their abilities are learned. Evidence that learning is conceptually linked with thinking is that the attribution of instinct, a denial of learning, to a particular animal's behavior is invariably a disparagement of that animal's cognitive complexity. If we say, for example, that a bird builds its nest instinctively, we are denying that that process involves thought or deliberation.

Even if machines exhibit superior computational or even conversational skills, we do not assume that they *learned* to do what they do. In the case of humans (and many animals) we do assume that they learned their skills.

If learning is essential, or at least very important, to our attributions of thinking then the following would be true: If we discovered of another human

that she or he had not grown up and learned like we had, but instead had been "built" fully grown by clever biologists who had created neural connections which allowed this creature to behave like a normal adult, without having to first learn such behavior, then we should be hesitant about whether this creature could actually think. On the other hand, knowing a robot were built so that it had to learn to walk and talk like a human infant does, would be reason to be less hesitant about believing this robot could think.

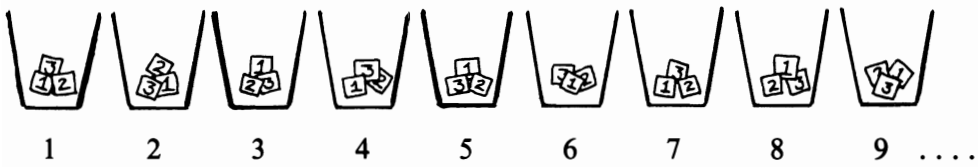
On this view, abilities equal or superior to humans would not necessarily be reason to assume that a creature could think. Those abilities—or at least some of them—would have to be learned.

However, there are conceptual difficulties with the idea of machine learning. Some students argued that machines could not learn because they can only do what they are programmed to do. What is meant by "programming" here is incompatible with learning. The word is often used in this sense—for example, by people who offer to "deprogram" the followers of certain religious sects, and by lawyers who hope to excuse their clients' deeds by convincing the juries that the clients were programmed to do what they did. We questioned whether this sense of "programming" was synonymous with the sense which means providing a set of instructions for a machine. We used a classroom demonstration and then a second encounter with the computer to show that there are ways of providing instructions to a machine which are not incompatible with the machine learning.

The classroom demonstration used a simple machine and a simple game called "Last One Loses" (a simple version of Nim) based on a paper by H. D. Block (1965). Last One Loses is a two-person, zero-sum, finite-state game played with a row of sticks set out on a table at the beginning of the game. Each player must take from one to three sticks at a turn. The one who takes the last stick loses. It is obvious to a human player that to win one must leave only one stick for the opponent's final turn. Then the opponent will be forced to take the last one.

The machine for this game can be made out of paper cups, as many as there are sticks on the table. Initially, in each cup there are three pieces of paper, numbered 1, 2, and 3. These represent the possible number of sticks that can be removed by the machine from each position. (See Figure 1)

Figure 1



The rest of the machine is simulated by a class participant. She follows the procedure below in the order listed unless an instruction directs one to a different step:

A. A piece of paper is drawn at random from the cup which corresponds to the number of sticks on the table.
B. If the number on the paper is greater than the number of sticks remaining, go to MACHINE LOSES.
C. If the number on the paper is less than or equal to the number of sticks remaining, remove the appropriate number of sticks from the table.
D. If the number of sticks remaining is zero, go to MACHINE LOSES.
E. The opponent moves. After opponent's move, if the number of sticks remaining is zero, go to MACHINE WINS.
F. Count number of sticks remaining on table and locate appropriate cup. If cup is empty, go to MACHINE LOSES.
G. Replace paper drawn on previous move and go to Step A.

MACHINE WINS: The piece of paper drawn from the cup on the previous move is replaced in that cup.

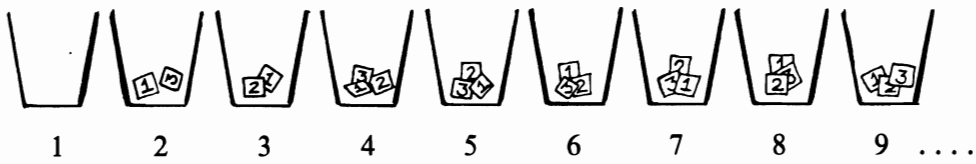
MACHINE LOSES: The piece of paper drawn from the cup on the previous move is discarded.

This cycle should continue until it becomes clear from an omniscient observation of the contents of the cups that no more papers will be discarded, i.e., that the machine has reached a steady state.

Block's paper cup machine is equipped with the ability to make the moves of Last One Loses. It can choose from one to three sticks at each turn. It can also keep track of how many sticks are left and it can eliminate its own bad moves. But it is not *programmed* to make any particular move; it is not programmed to win. Its program merely discards the last move it made preceding a defeat. As a result of playing the game repeatedly, its performance converges on the optimal strategy.

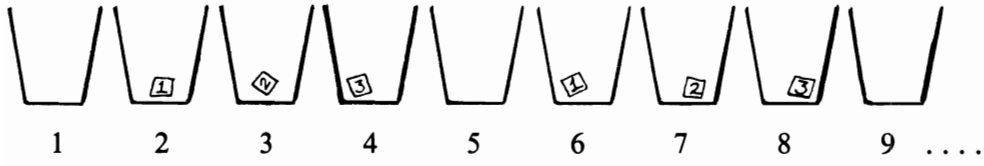
After several rounds with the class playing against the machine, all the papers in the first cup had been eliminated because any move the machine made when there was only one stick left was a losing move. (See Figure 2)

Figure 2



In effect, the machine resigns when it is forced to select a move from the first cup. After this point, the machine discarded any move that resulted in its having to draw from the first cup in the next turn. The better the machine's opponent, the faster it reaches the optimal condition for playing the game. (See Figure 3)

Figure 3



Although the machine initially moved at random, it was able to use the information gained from playing the game to improve. The program did not determine the final state of the paper cups, the experience with the game did that. If the machine had played a different game, such as Last One Wins, the final state of the paper cups would be different although the program and construction of the machine would be unchanged.

As the game was being played in class, attention was drawn to a similarity between the machine's learning and some other cases of learning. A graph of the games was made, where the x axis was the number of games played and the y axis was the number of wins by the machine. The graph looked like a typical cumulative learning curve for laboratory animals (including humans). There was an initial horizontal trend which began a sporadic and then accelerating upward climb, until the machine was winning every game possible.[5]

Class participants suggested differences between what the machine did and human learning: the machine could not make mistakes after it learned, the machine did not use strategies and plan ahead, the machine could not refuse to learn, the machine could not choose to lose, the machine could not apply what it learned in this case to new situations. It was pointed out that not all human (and other animal) learning was done deliberately, with awareness, or even voluntarily: rats learn to run mazes (rats with maze experience learn to navigate a new maze faster than inexperienced rats); wood ducks learn to recognize the wood duck call; and humans learn to speak with regional accents they would prefer not to have.

Suggestions were made about how the program could be altered to permit the machine to make mistakes, change strategies, apply what it learned to a new situation. For example, variations in the change made to the contents of the cups could include: 1) adding papers with numbers of the winning moves (positive reinforcement alone—a slow process, but it would not be deterministic and the machine would always be able to make mistakes); 2) adding papers with numbers other than the losing number, reinforcing alternative behaviors (even slower); 3) using a series of different methods of adding and discarding of papers, to be tried at random if the machine does not begin winning after a time. The most successful method could then be used for playing other games (such as Even Wins, see below).

Although the simplicity of the Paper Cup Machine and Last One Loses had heuristic advantages, a human was involved in the process (the student simulator), and the machine was slower to learn than the class. So the class

again used the computer. This time they played a slightly more complicated game whose winning strategy was not immediately obvious: Even Wins. Even Wins is played with an odd number of sticks; players must take from one to four sticks at a turn, and the player who has an even number of sticks when all have been taken is the winner. Programs were written by three members of the class,[6] based on the principles of the Paper Cup Machine. The operations previously simulated by a human were built into the program this time. Although the program at first had problems such as choosing to take more sticks than there were left, after a while it began to win. Class participants found themselves learning how to play by watching the moves made by the computer. In one program, after playing the game for a while, the computer was able to concede by typing "You win" when it was in a position from which all its moves had been discarded. Concessions early in a round when there were still many sticks left on the board had a one-upmanship effect which delighted the machine partisans in the class.

The Paper Cup Machine and second computer session illustrated that a machine can be programmed to improve its abilities on the basis of information received from its enviornment—a change similar to what, in most living organisms, we would call learning. The experiences gave many class participants a new conception of programming. Now they understood how a computer could do something that was not entirely due to the human programmer. The people who had written the programs *did not even know* the best moves for Even Wins and watched the computer's improvements like proud parents.

IV. *Computer Models in Research*

The demonstration of learning machines also served to introduce another important issue for class discussion: the role of computer models in scientific research. The similarity in learning curves for humans and the Last One Loses machine was offered as evidence that the machine, at least in some rudimentary sense, learns. But we can reverse our perspective, and ask instead, "Just what does the similarity between the human and the machine tell us about human learning?" If we view the computer as a model of human learning, how much can we learn from the model about the thing being modeled?

Many psychologists and researchers in artificial intelligence are working on computer programs, claiming to simulate human cognitive processes. Their claim is that we can learn something about how humans solve problems, recognize patterns, etc., by studying how a computer produces similar results.

Criticisms of such claims are frequently heard. It is certainly true that similarity of results, alone, does not establish similarity of processes. The fact that a computer can produce the same answer to a problem as can a human, tells us nothing, in itself, about the processes the human used to arrive at the answer. Yet, the differences often pointed out between humans and machines (say, that one contains neurons while the other contains transistors) in no way establish that a computer will be useless in the study of cognitive processes. Dif-

ferences or similarities of one feature may be quite independent of differences or similarities in other features. Many of the criticisms of models miss the point modelers are attempting to make. Model builders are not claiming that their models are similar in all respects to the system being modeled—they are only claiming that the two are similar in the relevant respects.

The problem, then, in evaluating cognitive (or other) models is to determine which similarities or differences between the model and the thing being modeled can be taken as evidence for similarities or differences of other features—specifically, those other features of the system being modeled that we would like to know more about. Students can discuss the reasons for thinking that certain features of a computer model (the order in which it performs operations, for example) are more more important than others (the hardware used to perform those operations) if we are interested in learning something about human cognitive processes.

The value of the use of models to suggest experiments can also be discussed. One way to decide whether a particular sort of similarity holds between a human and a machine, in a situation where we know what the machine is like, but are unsure what the human is like, is to devise an experiment that will give us more information about the human. Experimental results, of course, are subject to problems of interpretation, but the point to be made is that models often raise useful, and previously unasked, questions.

The question of which similarities or differences between a model and the thing being modeled are the important ones in a given context is a difficult, and to a large extent unanswered, question. Discussion of the problem can make students aware of the shortcomings of the simple answers often suggested to such a question, and introduce them to some of the complexities of scientific reasoning.

V

Setting up the two lab sessions we have described required very little experience with computers.[7] To get computer time for the course, we only needed to fill out a form describing what we wanted to do and submit it to the computer science center. Computer time costs money and although we were offered some financial assistance from the philosophy department, we applied timidly for one thousand dollars of computer time. Our request was immediately answered with a grant of *six* thousand dollars worth of computer time. Apparently the computer science center, which is attached to the computer science department, welcomed the opportunity to attract philosophy students to their program. (Perhaps departments of computer science at other schools need to be told of this opportunity.) A detailed manual from the center provided the information we needed for establishing separate account numbers and passwords for each student. The ELIZA program was in the computer library so we only needed the commands required to run it.

As for the learning program, we simply relied on students in the class.

Writing the program and getting it running was offered as an extra credit assignment. Our class of 50 students, only half of whom had ever operated a computer terminal before, produced three working programs and a few that almost worked.

If the ELIZA program is not in your computer library and/or you have no amateur programmers to write programs for learning Even Wins, throw yourself on the mercy of the computer facility staff. They may have to order a tape, type in the list of program statements, or even write a small program for learning Even Wins.[8] Whatever is needed, university computer facilities are almost always staffed by enthusiastic people who love to solve problems about computers and win converts. They are more likely to want to write a program than a letter requesting a tape.

We reserved a room with eight terminals for two meetings of each discussion section. An introduction to the programs we would be using had been provided in lecture beforehand. At the section meetings, it was necessary only to hand out account numbers and tell students what commands should be typed to gain access to the programs. Students who had never used the computer were paired with experienced students so that they could learn how to sign on and off the machine. Students who wanted more time with the computer (and most did) could return at their convenience, since each student had a personal account number.

Obviously there will be variations among computer facilities, and not everyone wishing to teach such a course will be able to set it up exactly the way we did. What we would like to stress, though, is that very little prior familiarity or experience with computers is required to run the lab sessions successfully. We relied, for the most part, on the knowledge and enthusiasm of a number of our students. It is likely, also, that many computer centers or departments have someone who can provide the necessary help.

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Weizenbaum (1976) and Raphael (1976) were the main texts used in this section of the course along with Anderson's anthology (1964). They both give a description of the basic principles of computers for a general audience, and discuss some of the applications in artificial intelligence, including the ELIZA program. Weizenbaum digresses frequently to warn the reader against an uncritical awe of science and computer abilities; Raphael's book is easier to follow. The books by Apter (1970) and Arbib (1972) overlap Weizenbaum and Raphael somewhat. Apter has more discussions on philosophy of mind and Arbib has an excellent more advanced discussion of brain modelling. Block and Ginsberg (1968) give a popular summary of developments in robotics, and the section from Winograd (1976) summarizes developments with computers and natural language.

Turing's (1950) article is, of course, about the question "Can machines think?" and the Imitation Game. Gunderson's (1971) chapter is a criticism of Turing. Both of these are in Anderson (1964). Both Putnam (1964) and Asimov (1970) relate philosophy of mind issues to machines; Putnam does entertaining (but thorough) philosophy and Asimov does philosophical entertainment.

Block's (1965) paper is a description of the machine that learns to play Last one Loses and Even Wins.

Chapter 4 of Fodor (1968) is an excellent introduction to the questions raised by the use of computer models in psychology. The paper by Newell, Shaw, and Simon (1958) is an early example of the use of the computer simulation approach in psychology, which contains an interesting discussion of why they have chosen such an approach, and how their approach is related to other approaches to psychology.

Zeigler (1976) is a comprehensive formal development of models, including a discussion of those cases in which valid inferences can be made about the thing being modeled based only on the formal properties of the model (Chapter 15). Suppe's

(forthcoming) review of Ziegler suggests some ways in which Ziegler's account of model validity might be extended.

A discussion of why it is important for philosophers of science to examine actual reasoning patterns in science can be found in Shapere (1974) and part III of the "Afterword" of Suppe (1977).

Notes

1. Adapted by H. D. Block from a limerick by W. R. Ashby.

2. Course participants were told to work at getting a realistic and intelligible conversation, inventing a problem if necessary. Otherwise ELIZA'S responses would be less plausible and less interesting.

3. Arbib, *The Metaphorical Brain*, Wiley Interscience, New York (1972): 87.

4. *Ibid.*

5. That is, winning when the machine's first move was from a position in which there was a winning strategy: where the number of sticks was *not* $4x + 1$ (where x is any integer).

6. The successful programs for learning Even Wins were produced by Bill Davis, Eric Lunow and Rick Thomas. We would like to thank them for this extra work and for their being the kind of students that make their teachers want to do extra work too.

7. The course was taught at the University of Maryland, College Park, in 1977.

8. A program for Even Wins is now available in David Ahl's *Basic Computer Games* (Morristown, NJ: Creative Computing, 1978): 60-2.

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